7 **(a)** State what is meant by a fundamental particle.

[1]

(b) A nucleus X has 14 nucleons and *p* protons. The ratio of charge to mass for nucleus X is $4.1 \times 10^{7}\,\mathrm{C\,kg^{-1}}$.

(i) Determine *p*.

[3]

(ii) Nucleus X undergoes β^- decay to form nucleus Z.

Complete the equation representing this decay.

$$^{14}_{\cdots\cdots}X \rightarrow {}^{\cdots\cdots}_{\cdots\cdots}Z + {}^{\cdots\cdots}_{\cdots\cdots}\cdots\cdots + {}^{\cdots\cdots}_{\cdots\cdots}\cdots\cdots$$

[3]

(c) A sample of a radioactive substance emits particles that are positively charged and have a continuous range of kinetic energies.

State and explain whether the nuclei in the sample are undergoing α-decay, β^+ decay or β^- decay.

[2]

[Total: 9]